Computer Networking

Inter - VLAN

Class Material



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v1.0

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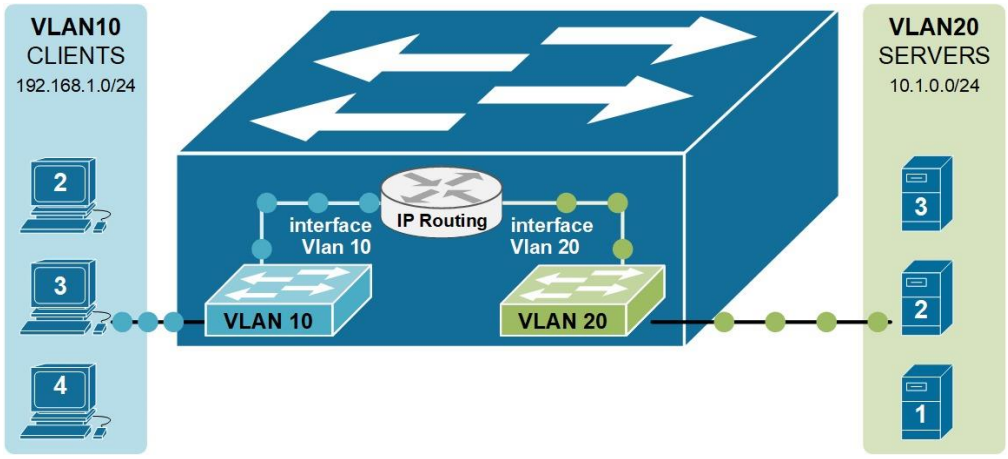
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1| What is Inter Vlan

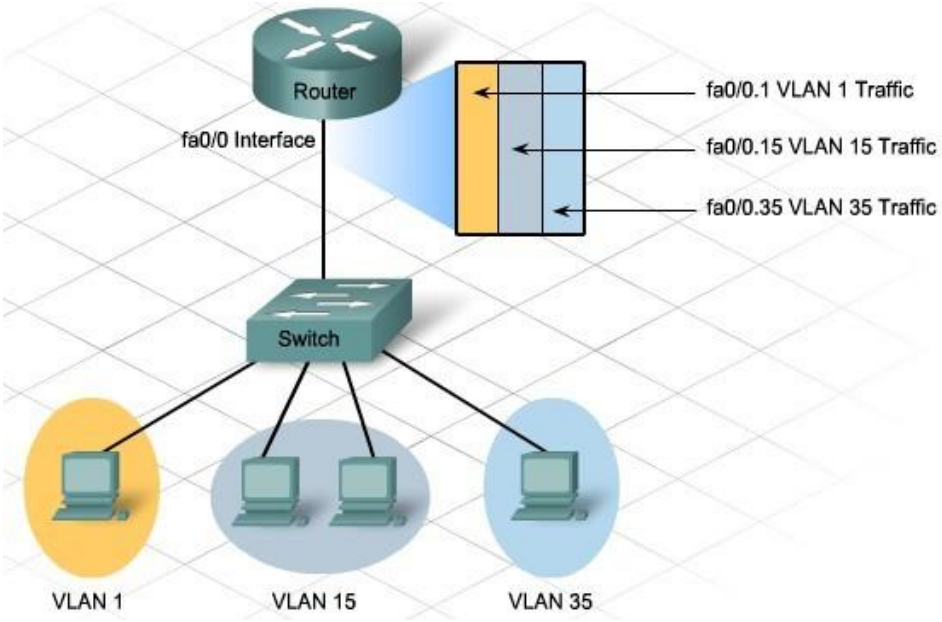
Inter-VLAN routing is the process of enabling communication between different Virtual Local Area Networks (VLANs). While VLANs isolate traffic to improve security and reduce broadcast domains, they prevent direct communication between subnets. Inter-VLAN routing breaks this isolation using Layer 3 devices (routers or multilayer switches) to move packets across VLAN.

Since VLANs are separate networks, they cannot communicate directly with each other.So, if devices in VLAN 10 want to talk to devices in VLAN 20, Inter-VLAN Routing is needed — that is, using a router or Layer 3 switch to send data between VLANs.



Why Is Inter-VLAN Routing Important?

Modern enterprise networks require seamless communication to support various business functions. Inter-VLAN routing ensures employees across different departments, applications, and workflows can collaborate effectively without compromising the integrity or security of the network.



Configure the Switch

Switch(config)#**vlan 5**

Switch(configvlan)#**exit**

Switch(config)#**vlan 10**

Switch(configvlan)#**exit**

Switch(config)#**vlan 15**

Switch(configvlan)#**exit**

Switch(config)#**interface GigabitEthernet 1/1**

Switch(configif)#**switchport mode trunk**

Switch(configif)#**exit**

Configure the Router

- *For the **VLAN 10**:*
  - `Router(config)#interface ge0/1.10`
  - `Router(configsubif)#encapsulation dot1q 10`
  - `Router(configsubif)#ip address 192.168.10.1 255.255.255.0`
  - `Router(configif)#no shutdown`
- *For the **VLAN 15:***
  - `Router(config)#interface ge0/1.15`
  - `Router(configsubif)#encapsulation dot1q 15`
  - `Router(configsubif)#ip address 192.168.15.1 255.255.255.0`
  - `Router(configif)#no shutdown`
- Verify configuration with the commands:
  - `Router#show vlans`
  - `Router#show ip route`
- Test configuration with **ping** and **traceroute**

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